MIL-E-13500B(ME)
26 October 1978
SUPERSEDING
MIL-E-13005A(ME)
2 April 1973

MILITARY SPECIFICATION

EXPOSURE CONTROL, PHOTOLITHOGRAPHIC, LIGHT INTEGRATING

This specification is approved for use by the Mobility Equipment Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers light-integrating type exposure controls used primarily to control the exposures of lithographic plates and photographic negatives.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

FSC 6760

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

FEDERAL

QQ-S-781	- Strapping, Steel, and Seals.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-B-640	- Boxes, Fiberboard, Corrugated, Triple-Wall.
PPP-C-843	- Cushioning Material, Cellulosic.

MILITARY

MIL-P-116	- Preservation-Packaging, Methods of.
MIL-P-514	- Plates, Identification, Instruction and Marking, Blank.
MIL-V-173	- Varnish, Moisture-and-Fungus-Resistant (for Treatment of Communications, Electronic, and Associated Equipment).

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-461	- Electromagnetic Interference Characteristics Requirements for Equipment.
MIL-STD-1188	- Commercial Packaging of Supplies and Equipment.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated the issue in effect on date of invitation for bids or request for proposal shall apply.

SOUTHERN CALIFORNIA AIR POLLUTION CONTROL DISTRICT

Rule 102 - Photochemically Reactive Solvents.

(Application for copies should be addressed to the Southern California Air Pollution Control District, 434 South San Pedro Street, Los Angeles, CA 90013.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Description. The light integrating photolithographic exposure control (hereinafter called "exposure control") shall consist essentially of a main control unit and a photoelectric cell unit, with all necessary connectors and cables required for proper operation.

3.2 First article (first produced exposure control). The contractor shall furnish one or more exposure controls as specified (see 6.2) for examination and testing within the time frame specified (see 6.2) to prove that his production methods and choice of design detail will produce exposure controls that comply with the requirements of this specification. Examination and tests shall be as specified in Section 4 and shall be subject to surveillance and approval by the Government (see 6.3).

3.3 Material. Material shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification (see 6.6).

3.4 Main control unit. The main control unit shall be provided with an "on-off" line switch or button with pilot light to indicate when power is "on", a switch or button to initiate the exposure, a means of instantly cancelling an exposure, and an elapsed time indicating dial to manually set various exposure durations. Provisions shall be made on the instrument to make repeated identical exposures.

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The power cable for the unit shall be not less than 10-feet (3.048 meters (m)) long and shall be equipped with a three prong, grounded, male plug. The main control unit shall be housed in a case having a cubage not more than 250 cubic inches (.0041 m^3) and no dimension longer than 7 inches (0.178 m). Lugs with drilled holes or key-hole slots shall be provided on the outside of the case to permit mounting the unit on a wall. All controls switches, and electrical plugs shall be labeled. Two spare bulbs shall be provided for the pilot light.

3.4.1 Indicating dial. The indicating dial shall be graduated in intervals of not less than 1000th of the full range of the equipment.

3.5 Photoelectric cell. The photoelectric cell shall be a photo diode type. The case shall have a window of light diffusing glass through which all light must pass to reach the photoelectric cell. Two spare light diffusing glasses shall be furnished. The cell shall be housed in a dustproof metal case having a cubage of not more than 6 cubic inches (.0001 m^3) and with the shorter dimension on the window side of the case being not more than 1.5 inches (0.038 m). A cable not less than 25-feet (7.620 m) long and equipped with a suitable quick-connect type plug for connection to the main control unit shall be provided.

3.5.1 Photoelectric cell mounting. The photoelectric cell case shall be provided with a mounting bracket so the photoelectric cell may be removed without tools. The mounting bracket shall be provided with straps with drilled and countersunk holes or key hole sockets positioned to permit permanent mounting of the unit on the front of a printing frame or camera.

3.6 Remote control unit. The remote control unit shall be equipped with a switch to initiate the exposure and an auxiliary "on-off" switch to activate the shutter. All controls shall be labeled. A power cable at least 10-feet (3.048 m) long, with a quick-connect type plug, shall be provided for connection to the main control unit. The unit shall be housed in a dustproof case having a cubage not more than 43 cubic inches (.0007 m^3), and no dimension longer than 7 inches (0.178 m). Metal straps with drilled holes or keyhole slots shall be provided on the outside of the case to permit mounting the unit on a wall.

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3.7 Range. The exposure control shall be a multi-range instrument with a basic range and one or more secondary ranges. When the dial is set at the maximum range position on the basic range scale and the photoelectric cell window is illuminated with 1,000 foot candles, an exposure time of 200 seconds, plus or minus 10 percent, shall result. The secondary range or ranges shall provide an exposure duration of at least 10 minutes. The different ranges may be obtained by the use of a scale multiplier switch, filters, metal Waterhouse stops, or combinations thereof. If filters are used, they shall not bleach nor darken through aging or exposure to light.

3.8 Sensitivity and stability. The exposure control shall be stable within a time range from 10 seconds to 10 minutes and voltage range from 95 to 125 volts with a light intensity of 1,000 foot candles on the photoelectric cell.

3.9 Electrical requirements. The exposure control shall operate on 95-125-volt, 50-60-Hertz (Hz), single-phase current and shall contain all transistorized circuitry. The instrument shall be protected with a properly rated replaceable type fuse. Five spare fuses shall be furnished with each instrument. All components of the exposure control shall be grounded.

3.10 Electromagnetic interference. When specified (see 6.2), the electromagnetic interference emission and susceptibility characteristics of the exposure control shall conform to MIL-STD-461, Class II B.

3.11 Fungus and moisture resistance. The electrical circuitry, including all components and connections, except as specified below, shall be protected from the effects of fungus growth and moisture by an overall treatment with a varnish conforming to MIL-V-173, Composition I or II, as applicable (see 3.11.1), with 1 percent copper 8-quinolinolate (by weight) based on the nonvolatile content of the varnish.

(a) Components or circuit elements that are inherently fungus and moisture resistant or which are hermetically sealed need not be treated.

(b) Components or circuit elements whose functions will be adversely affected by the varnish coating shall not be treated.

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When used, the varnish shall be applied by spray, brush, or a combination of both to give a dry-film thickness of not less than 1 mil to component or element surfaces previously cleaned and prepared so that the surfaces are free from all foreign matter which would interfere with the adherence or function of the varnish.

3.11.1 Composition. Composition II shall be used only in the case when local air pollution regulations governing the application of varnish precludes the use of Composition I. When Composition II is used, the contractor shall provide evidence to the Government that the use of Composition II is required, and shall certify that the Composition II material complies with Rule 102, Southern California Air Pollution Control District.

3.12 Identification marking. The exposure control shall be identified in accordance with MIL-STD-130. The marking shall be applied to the exposure control on plates conforming to MIL-P-514, Type I, Style 1, Composition C, of Type I, Grade A, Class 1 material. The plates shall be attached by screws, bolts, or rivets in a conspicuous protected location.

3.13 Treatment and painting. The exposure control shall be cleaned, treated, and painted in accordance with the manufacturer's standard practice for this type of equipment.

3.14 Workmanship. The exposure control shall be free from defects that could impair its operation, accuracy, or serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards, as applicable.

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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First produced exposure control inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).
(c) Inspection of packaging (see 4.6).

4.3 First produced exposure control inspection.

4.3.1 Examination. The first produced exposure controls shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The first produced exposure controls shall be tested as specified in 4.5.2.1 through 4.5.2.7. Failure of any test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination and tests shall be in accordance with MIL-STD-105, Inspection Level III.

4.4.2 Examination. Samples selected in accordance with 4.4.1 shall be examined as specified in 4.5.1. AQL shall be 2.5 percent defective for major defects and 4.5 percent defective for minor defects.

4.4.3 Tests. Samples selected in accordance with 4.4.1 shall be tested as specified in 4.5.2.1 through 4.5.2.6. AQL shall be 2.5 percent defective.

4.5 Inspection procedure.

4.5.1 Examination. The exposure control shall be examined as specified herein for the following defects:

Major

101. Material not as specified.
102. Components missing or not as specified.
103. Electromagnetic interference suppression components and network not identical to the approved model.
104. Electrical circuitry not protected from the effects of moisture and fungus growth.

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105. Identification marking missing or illegible.
106. Workmanship not as specified.

Minor

201. Treatment and painting not as specified.

4.5.2 Tests.

4.5.2.1 Test conditions. Tests shall be conducted with the constant light source specified in 4.5.2.5 and the lightproof box specified in 4.5.2.6. The length of each exposure of the exposure control shall be timed with an accuracy of plus or minus 0.5 percent for exposures over 1 minute. The duration of exposure (T) shall be determined by averaging 5 timed exposures. For each duration of exposure (T), the constant (K) shall be calculated by means of the following formula:

$$K = \frac{SD^2}{TF}$$

Where:

K = Constant
S = Setting of dial
T = Duration of exposure (seconds)
D = Distance between photoelectric cell and light source (plus or minus 0.050 inch) (plus or minus 1.3 mm).
F = Percent transmission, neutral density filter (plus or minus 0.04 density units).

4.5.2.2 Dial. With the light source 24 inches (0.61 m) from the photoelectric cell the duration of exposure (T) shall be determined when the dial is set at 10-, 20-, 40-, 60-, 80-, and 100-percent of full scale. If the exposure control is equipped with a scale multiplier switch or a scale selector switch, then at least one duration of exposure (T) shall be determined while the switch is in each available position. Deviation of K value from the average for any dial setting of more than plus or minus 5 percent shall constitute failure of this test.

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4.5.2.3 Variable intensity. The dial shall be set at either 20- or 40-percent of full scale and the duration of exposure (T) determined with the light source set at 12, 18, 24, 30, and 36 inches (0.31, 0.46, 0.61, 0.76, and 0.91 m) from the photoelectric cell. A duration of exposure (T) shall also be determined with the light source and dial set in such positions as to result in a length of exposure of at least 10 minutes. If an exposure of this length cannot be achieved even when the dial is at full scale of the longest scale and the light source is at the 36 inch (0.91 m) position, then an appropriate filter may be used. Deviation of K value from the average for any distance setting more than plus or minus 5 percent shall constitute failure of this test.

4.5.2.4 Voltage. The light source and the dial shall be set in such a position as to result in a length of exposure of between 60 and 70 seconds. The duration of exposure (T) shall be determined with voltages of 95-, 105-, 115-, and 125-volts. Deviation of K value from the average for any voltage more than plus or minus 5 percent shall constitute failure of this test. No manual adjustments shall be made to the instrument either prior to or during this test to compensate for various voltages.

4.5.2.5 Constant light source. The constant light source shall be a new 200-watt, 120-volt, incandescent frosted or opal lamp. The lamp shall be operated at a voltage of 100, plus or minus 5 volts. However, throughout the tests, the current value shall remain within 8 milli-amperes of its value at the start of the tests. The lamp shall be conditioned by allowing it to burn for 1 hour at its rated voltage before the tests are started. All tests shall be done over one continuous period without turning off the lamp.

4.5.2.6 Lightproof box. The lightproof box shall exclude all light other than the light emanating from the constant light source. The box shall be square and shall be long enough to conduct the tests described in 4.5.2.3. An opening large enough to fit the window on the photo-electric cell unit shall be made in one end of the box. The lamp shall be mounted on a sliding platform. Three baffles, with round holes approximately 50-percent larger in area than the photoelectric cell window, but in no case smaller than 1.5 inches (0.038 m) in diameter shall be equally spaced between the lamp and the photoelectric cell.

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The center of the photoelectric cell window, the holes in the baffles, and the center of illumination of the lamp shall be on a common axis which coincides with the center line of the box, throughout the range of movement of the lamp. All inside surfaces of the box and the baffles shall be painted with dull-black, nonreflecting paint. The box shall be ventilated so that the temperature rise inside the box at the photoelectric cell window will not be more than plus 10° F (5.56° C) above ambient room temperature.

4.5.2.7 Electromagnetic interference. When specified herein, the first produced exposure control shall be tested to determine conformance to 3.10. The contractor shall furnish the contracting officer a report of tests required by MIL-STD-461. Disapproval of the report shall constitute failure of this test.

4.6 Inspection of packaging.

4.6.1 Quality conformance inspection of pack.

4.6.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.6.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.1.3 Examination. Samples selected in accordance with 4.6.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

107.	Materials, methods, or containers not as specified for Level A or Level B. Each incorrect material, method, or container shall be considered one defect.
108.	Components not cushioned within the inner container as specified for Level A or Level B.
109.	Desiccant not placed within the inner container as specified for Level A.
110.	Quantities packed together have a weight in excess of 200 pounds.
111.	Box lids not secured with wood screws as specified for Level A.
112.	Strapping not as specified.
113.	Marking missing, illegible, incorrect, or incomplete.

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5. PACKAGING

5.1 Preservation. Preservation shall be Level A, Level B, or Commercial, as specified (see 6.2).

5.1.1 Level A. Components comprising a complete exposure control shall be preserved in accordance with MIL-P-116, Method IIb, using boxes conforming to PPP-B-636. The inner container shall be a Style RSC, Type CP, Class Domestic, Variety SW, Grade 125, and the outer container shall be a W6c. The desiccant shall be placed in the inner container and the components in the container cushioned, to prevent movement and damage, with cushioning material conforming to PPP-C-843, Type II, Class B. Outer container closure shall be in accordance with the appendix of the box specification, Method V.

5.1.2 Level B. Components comprising a complete exposure control shall be preserved as specified in 5.1.1 for Level A except the method of preservation shall be IA-14 and closure of the outer container need not be Method V.

5.1.3 Commercial. Each exposure control shall be preserved in accordance with MIL-STD-1188.

5.2 Packing. Packing shall be Level A, Level B, or Commercial, as specified (see 6.2).

5.2.1 Level A. Exposure controls, preserved as specified in 5.1, shall be packed in quantities not to exceed 200 pounds in close-fitting boxes conforming to PPP-B-601, Overseas Type, style optional. Strapping shall conform to QQ-S-781, Class 1, Type I or IV, Finish A, size as applicable. (Note: When exposure controls are preserved Level A, the box lids shall be secured with wood screws instead of nailing.)

5.2.2 Level B. Exposure controls, preserved as specified in 5.1, shall be packed as specified in 5.2.1 for Level A except that boxes shall conform to PPP-B-601, Domestic Type, or PPP-B-640, Class 2, style optional.

5.2.3 Commercial. Exposure controls shall be packed in accordance with MIL-STD-1188.

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5.3 Marking.

5.3.1 Military packaging. Marking shall be in accordance with MIL-STD-129.

5.3.2 Commercial packaging. Marking shall be in accordance with MIL-STD-1188.

6. NOTES

6.1 Intended use. The exposure control is used to automatically control the quantity of light required to expose lithographic plates or photographic negatives regardless of fluctuations in the light source.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Time frame required for submission of first produced exposure control and number of items required (see 3.2).
(c) When electromagnetic interference compatibility is required (see 3.10).
(d) Degree of preservation and degree of packing required (see 5.1 and 5.2).

6.3 First produced exposure control. Any changes or deviations of production exposure controls from the approved first produced exposure control during production will be subject to the approval of the contracting officer. Approval of the first produced exposure control will not relieve the contractor of his obligation to furnish exposure controls conforming to this specification.

6.4 Data requirements. The contracting officer should include requirements for such data as technical publications, instructional materials, illustrated parts lists, and contractor's maintenance and operation manual to be furnished with each exposure control.

6.5 Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools, as necessary (including any special tools), and instructions regarding shipment of exposure controls.

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6.6 Recycled material. It is encouraged that recycled material be used, when practical, as long as it meets the requirements of this specification (see 3.3).

Custodian:
Army - ME

Preparing activity:
Army - ME

Review activity:
DLA - GS

Project 6760-A553

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE: MIL-E-13500B(ME) Exposure Control, Photolithographic, Light Integrating

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO (If "Yes", in what way?)

4. REMARKS

SUBMITTED BY (Printed or typed name and address - Optional)

TELEPHONE NO.

DATE

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